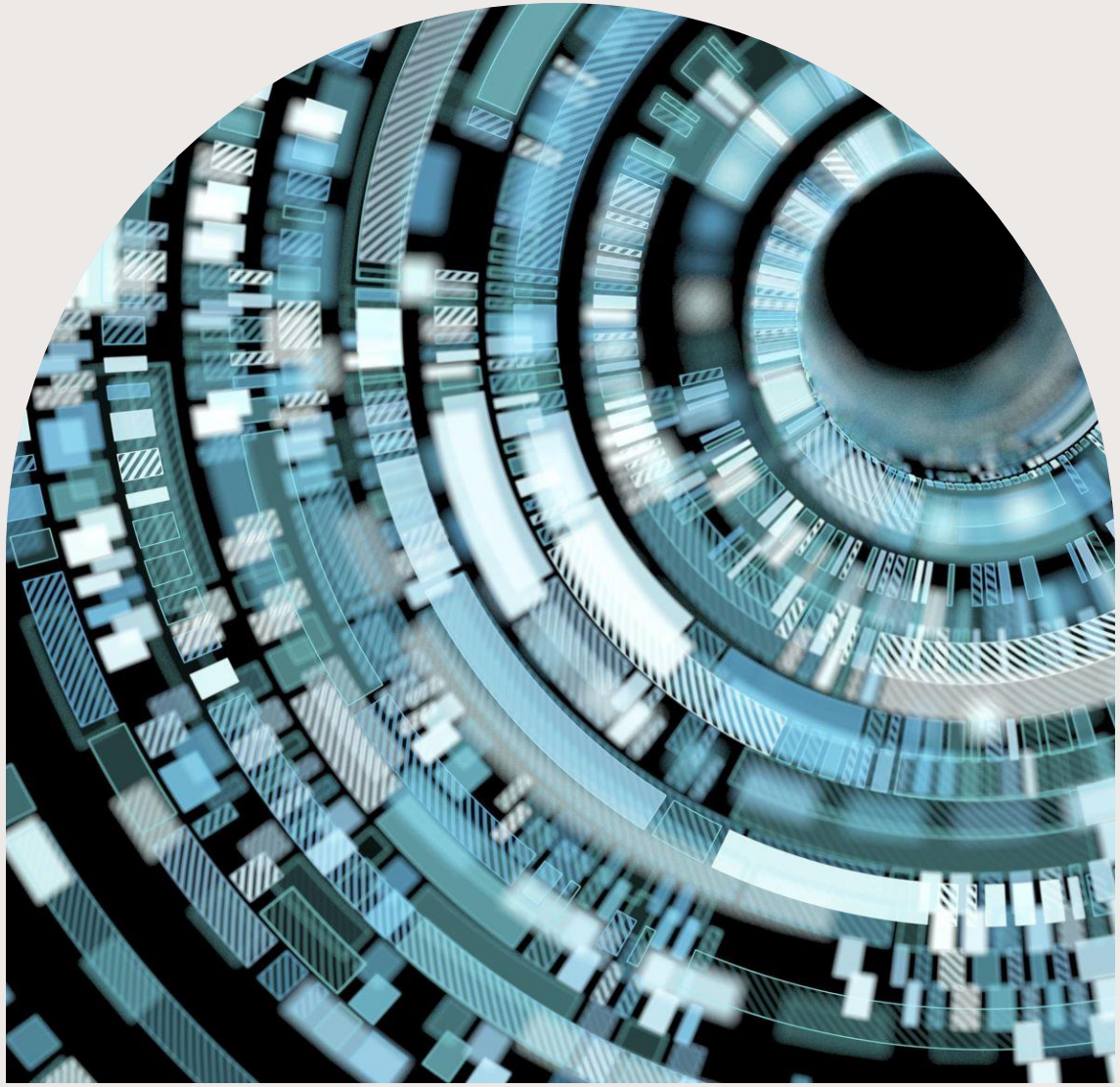


MACHINE LEARNING

Regularization: L1 & L2

INTRODUCTION

- It is a technique we use to overcome overfitting and depends on the type of algorithms that we are using. But, while primarily used for overfitting, regularization parameters are tuned via cross-validation to find the balance.
- For linear regression models – Adding a penalty term in the cost function
- For Decision tree models – Pruning of nodes, if deeper is the tree
- For Neural Networks – Drop out the neurons (randomly switching off the neurons)
- **For Linear regression models-:**
- Regularization Cost Functions take the form:
- Regularization Estimator, **Cost=Loss(RSS)+Penalty**

LASSO REGRESSION(L1 REGULARIZATION)

- Least Absolute Shrinkage and Selection Operator
- Adds a penalty proportional to the absolute value of the magnitude of coefficients:
- Lasso specifically targets the coefficients of irrelevant or less important predictors.
- Coefficients are shrunk toward zero, and with ahigh enough penalty (λ) they become exactly zero, effectively removing them from the model, *performing feature selection*
- Lasso Regression is the primary method for shrinking coefficients to exactly zero, performing automatic feature selection by applying an (L1)penalty to the model. It forces coefficients of less important, redundant, or noisy features to zero, creating a sparse, more interpretable model

$$\text{Cost} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{m} |w_j|$$

- Small λ (<0.0001): Weak regularization, might still lead to overfitting.
- Moderate λ (0.0001 – 0.01): Commonly recommended starting range.
- Large λ (>0.1): Strong regularization, might cause underfitting

RIDGE REGRESSION(L2 REGULARIZATION)

- Adds a penalty proportional to the square of the magnitude of coefficients:
- It will reduces model complexity and prevents overfitting by shrinking coefficients, but rarely tozero.
- Penalty terms used here are:
- λ(Lambda or Alpha) - Regularization(tuning)parameter, that decides how much you want topenalize the model's complexity.
- A higher λ increases penalty strength, reducing overfitting but potentially causing underfitting.
- wj - Model coefficients (weights) (coefficients of each feature/attribute available in the dataset)
- m : Number of features

$$\text{Cost} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{m} w_j^2$$

- Note that ridge regression does not shrink every coefficient by the same value. Rather, coefficients are shrunk inproportion to their initial size. As λ increases, high-value coefficients shrink at a greater rate than low-valuecoefficients. High-value coefficients are thus penalized greater than low-value coefficients.
- Ridge regression introduces a penalty ($L2$) to reduce variance by shrinking coefficients, which intentionally addsbias to the model. As λ approaches zero, this artificial bias is reduced, allowing the model to fit the training datamore closely.
- Ridge Regression (L2) shrinks coefficients toward zero but rarely sets them to exactly zero - It's great for dealing with*multicollinearity (when your variables are highly correlated)*.
- *Models regularized through ridge regression produce less accurate predictions on training data (higher bias) butmore accurate predictions on test data (lower variance). This is bias-variance trade off. Through ridge regression,users determine an acceptable loss in training accuracy (higher bias) in order to increase a given model'sgeneralization (lower variance). In this way, increasing bias can help improve overall model performance.*

SUMMARY

- Both lasso regression and ridge regression reduce model complexity, but in different ways.
- Lasso regression reduces the number of independent variables affecting the output.
- Ridge regression reduces the weight each independent variable has on the output.